MILITARY SPECIFICATION

MICROPHONE M-161( )/TNS-10

This specification is approved for use by the Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of microphone for detecting the sounds of artillery fire and shell detonation. The microphone is part of the Sound Ranging Set AN/TNS-10 and is designated as Microphone M-161( )/TNS-10 (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901	Shock Tests, H. I. (High Impact); Shipboard Machinery, Equipment and Systems Requirements for
MIL-S-10956	Sound Ranging Set AN/TNS-10( ), Packaging of
MIL-P-11268	Parts, Materials and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Command, ATTN: DRSEL-RD-TS-S, Fort Monmouth, New Jersey 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5965

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STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and Other Devices
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

SC-A-678032	Burn-in Procedure
DL-SM-B-690400	Microphone M-161( )/TNS-10
SC-A-883795	Performance Test Procedure for Microphone M-161( )/TNS-10
SC-A-883796	Environmental Test Procedure for Microphone M-161( )/TNS-10

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction. Microphone M-161( )/TNS-10 shall be constructed in accordance with this specification and Drawing and Data List DL SM-B-690400 including the requirements for parts, materials, and processes thereon.

3.2.1 Parts, materials, and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply.

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3.2.2 Systems safety engineering.

3.2.2.1 Personnel hazards. Personnel hazards shall be kept to a minimum through compliance of configuration changes and parts selection with Requirement 1 of MIL-STD-454 (see 4.10).

3.2.2.2 Edge rounding, exposed. All exposed edges and corners shall be rounded (see 4.10).

3.2.2.3 Radioactive materials. Radioactive materials shall not be used (e.g. self luminous dials or markings, electron tubes, surge arrestors and lenses) (see 4.10).

3.3 Performance requirements.

3.3.1 Dielectric strength. The input circuitry of each completely assembled microphone, with the lightning arrestors removed, shall withstand a 5 second application of 500 volts DC without degradation. (See 4.6.)

3.3.2 Wire grid assembly. The microphone wire grid assembly shall be constructed in accordance with the Drawing SM-D-883766 and meet the electrical requirements therein (see 4.6).

3.3.3 Frequency response. The frequency response of the microphone shall be within ± 3 dB for frequencies of 35 Hz and below and within ±4 dB for frequencies above 35 Hz of the nominal values shown in Table I below (see 4.6):

Table I. Frequency Response

Frequency (Hz)		Nominal Volts P-P	Max Volts P-P	Min Volts P-P
5	With	4.5	6.4	3.2
10	25 Hz	5.8	8.2	4.1
15	plugs	6.6	9.4	4.7
20		5.0	7.1	3.5
25		3.1	4.4	2.2
30		2.0	2.8	1.4
35		1.4	2.0	1.0
40		.9	1.4	.6
15		6.2	8.8	4.4
25		7.4	10.5	5.3
35	With	5.0	7.1	3.6
45	60 Hz	2.7	4.3	1.7
55	plugs	1.6	2.6	1.0
70		.9	1.4	.6
85		.6	1.0	.4

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3.3.4 Amplifier current. The collector current of the amplifier in the microphone shall decrease when positive sound pressure is impressed upon the microphone (see 4.6).

3.3.5 Amplifier frequency response. The microphone shall have a smooth frequency response curve over the frequency range of 5 to 300 Hz. The output at 15 Hz shall be 28 ± 0.5 volts (see 4.6).

3.3.6 Relay. The relay shall come full-on when 80 volts is applied to the signal terminals through a 5100 ±2 percent resistor (see 4.6).

3.3.7 Diode bridge. The performance of the microphone shall be unaffected by changes in power supply polarity (see 4.6).

3.4 Finish, protective. Equipment shall be given protective finish in accordance with MIL-F-14072. This includes finish of hardware, such as handles, hinges, screws, and necessary touch-up after mounting. The final paint film on type 1 surfaces shall be green color (olive drab), semigloss enamel matching color chip X24087 of FED-STD-595.

3.5 Marking.

3.5.1 General. Marking shall be in accordance with MIL-M-13231 (see 4.10).

3.5.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.3 Serial numbers. Each microphone shall be serial numbered.

3.6 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall conform to Requirement 7 of MIL-STD-454 (see 4.7).

3.7 Burn-in. The microphone shall meet the requirements of this specification after being subjected to the burn-in cycle of Drawing SC-A-678032 (see 4.8).

3.8 Service conditions. After a warmup period of 10 minutes the microphone shall meet the performance requirements of this specification when subjected to the following conditions:

3.8.1 Temperature.

3.8.1.1 Low. The microphone shall meet the frequency response requirement of 3.3.3 at a temperature of -40°C(-40°F) after being subjected to a non-operating temperature of -45.5°C)(-50°F) (see 4.9.1.1).

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3.8.1.2 High. The microphone shall meet the frequency response requirement of 3.3.3 at a temperature of 65.6°C (150°F) after being subjected to a non-operating temperature of 71.1°C (160°F) (see 4.9.1.2).

3.8.2 Humidity. The microphone shall meet the frequency response requirement of 3.3.3 and shall be free of corrosion when subjected to 94 ± 4 percent relative humidity (see 4.9.2).

3.8.3 Immersion. After immersion the microphone shall meet the frequency response requirement of 3.3.3 with 25 Hz plugs and shall be within 1.5 dB of the readings taken prior to immersion (see 4.9.3).

3.8.4 Bounce and shock. The microphone shall meet the requirements of Table I.

Table I. Bounce and Shock

Requirement	Insp Para	Performance after test
Bounce	4.9.4	Full specification performance Minor mechanical damage is permissible.
Shock; ballistic	4.9.5	Full specification performance Minor mechanical damage is permissible.

3.8.5 Fungus. The microphone shall be resistant to fungus growth when subjected to fungi under conditions favorable for their development (see 4.9.6).

3.9 Workmanship. The microphone shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs in MIL-P-11268:

General requirements for plastic material and plastic parts.
Wiring and cabling, including:
- Slack
- Protection
- Clearance
- Splicing
- Connections, general
- Grounding, general
- Shielding on wire and cable

Soldering and brazing
Cleaning of units
General requirements for securing of parts
Process for welding
Cleaning after welding
Controls
Self-tapping screws
Cleaning prior to welding

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Inspections covered by subsidiary documents (see 4.4).

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before packaging (see 4.5).

(2) Quality conformance inspection of packaging (see 4.11).

4.3 First article. Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor (see 3.1).

4.3.1 First article units. Unless otherwise specified in the contract, the contractor shall furnish nine first article units of the complete Microphone M-161( )/TNS-10.

4.3.2 First article inspection. This inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, burn-in as specified in 4.9, and the inspections specified for Group A and Group C (see Tables III and IV respectively). First article inspection shall be performed in the order shown in Table II.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery.

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Item	Where required
Parts, materials, and processes	3.2
Finish	3.4
Marking	3.5
Workmanship	3.9

Table II. First Article Inspection

Inspections	Req Para	Test Para	Units 1-3	Units 4-6	Units 7-9
1. Inspections covered by subsidiary documents 1/	3.4 3.5	4.4	Inspection to be performed on all units		
2. Burn-in 1/	3.7	4.8	Inspection to be performed on all units		
3. Group A inspection 1/	See Table III		Inspection to be performed on all units		
4. Interchangeability 1/	3.6	4.7	1		
5. Environmental 2/					
Low temperature	3.8.1.1	4.9.1.1	2		
High temperature	3.8.1.2	4.9.1.2	3		
Humidity	3.8.2	4.9.2	4		
Immersion	3.8.3	4.9.3		1	
Bounce	3.8.4	4.9.4		2	
Shock; ballistic	3.8.4	4.9.5		3	
Fungus	3.8.5	4.9.6			1

1/ Inspections 1 to 3 shall be performed, in the order shown, on all first article units before subjecting these units to any other inspection requirements.

2/ The numbers in the units column for inspections 4 and 5 specify the order of inspections for the indicated unit.

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4.5 Quality conformance inspection of equipment before packaging. Every equipment shall be subjected to burn-in (see 3.7) prior to conformance inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.10. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of burn-in shall be classified as a defect and recorded as such.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table III. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subjected to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQLs indicated in Table III.

Table III. Group A Inspection

Inspection or Test	Req Para	Insp Para	AQL	
			Major	Minor
Visual and Mechanical	3.9	4.10	1%	4%
Electrical				
Dielectric strength	3.3.1	4.6		
Wire grid assembly	3.3.2	4.6	1% for	1/
Frequency response	3.3.3	4.6	the group	
Amplifier current	3.3.4	4.6		
Amplifier frequency response	3.3.5	4.6		
Relay	3.3.6	4.6		
Diode bridge	3.3.7	4.6		

1/ All electrical defects are considered major

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in any order satisfactory to the Government.

4.5.2 Group B inspection. Not required.

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A inspection. The inspection shall consist of the inspections specified in Table IV. Samples shall be selected in accordance with 4.5.3.1.

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Table IV. Group C Inspection

Inspection	Req Para	Insp Para
Subgroup 1		
Interchangeability	3.6	4.7
Low temperature	3.8.1.1	4.9.1.1
High temperature	3.8.1.2	4.9.1.2
Humidity	3.8.2	4.9.2
Subgroup 2		
Immersion	3.8.3	4.9.3
Bounce	3.8.4	4.9.4
Shock; ballistic	3.8.4	4.9.5
Subgroup 3		
Fungus 1/ 2/	3.8.5	4.9.6

1/ The fungus test shall be performed only during first article testing and on the first production lot.

2/ The microphones shall be thoroughly washed, cleaned, dried, and refurbished after this inspection before proceeding with subsequent inspections.

4.5.3.1 Sampling for Group C inspection. Nine sample microphones shall be selected at random, three samples for each Subgroup in Table IV, from the first one hundred sixty units or fraction thereof produced. The fungus test shall be performed only on the first production lot. Thereafter, six units of the microphone shall be selected at random, three samples for subgroups 1 and 2 of Table IV from every one hundred sixty units or fraction of at least eighty units produced. If the fraction is less than eighty, the selection of Group C samples for the preceding test shall include units from the remaining fraction.

4.5.3.1.1 Sampling for inspection of solderless-terminal-lug connections. Five specimens of each combination of wire and terminal shall be prepared each week for inspections. A specimen is a one foot length of wire on which a terminal lug has been applied, using the same tools and procedures used in crimping the terminal lugs in production.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in any order which is satisfactory to the Government.

4.5.3.3 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

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4.5.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection, may be accepted on contract provided all damage is repaired and the sample units are resubjected to and pass group A inspection.

4.6 Performance tests. The performance tests of Microphone M-161( )/TNS-10 shall be performed as specified in the test procedure, Drawing SC-A-883795.

Requirement	Paragraph
Dielectric strength	3.3.1
Wire grid assembly	3.3.2
Frequency response	3.3.3
Amplifier current	3.3.4
Amplifier frequency response	3.3.5
Relay	3.3.6
Diode bridge	3.3.7

4.7 Inspection for interchangeability. The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirement of 3.6. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable subassemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

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4.8 Burn-in. All units submitted for acceptance shall be subjected to burn-in in accordance with Drawing SC-A-678032 (see 3.7).

4.9 Service conditions. The microphone shall be tested in accordance with the following conditions:

4.9.1 Temperature test.

4.9.1.1 Low. The M-161( )/TNS-10 shall be subjected to the test of Method 502.1, Procedure I, of MIL-STD-810 as specified on Drawing SC-A-883796 (see 3.8.1.1).

4.9.1.2 High. The M-161( )/TNS-10 shall be subjected to the test of Method 501.1, Procedure II, of MIL-STD-810 as specified on Drawing SC-A-883796 (see 3.8.1.2).

4.9.2 Humidity test. The M-161( )/TNS-10 shall be subjected to the test of Method 507.1, Procedure II, of MIL-STD-810 as specified on Drawing SC-A-883796 (see 3.8.2).

4.9.3 Immersion test. The M-161( )/TNS-10 shall meet the requirement of 3.8.3 after being immersed as specified on Drawing SC-A-883796.

4.9.4 Bounce test. The M-161( )/TNS-10 shall be subjected to the test of Method 514.2, Procedure XI, of MIL-STD-810 as specified on Drawing SC-A-883796 (see 3.8.4).

4.9.5 Shock test; ballistic. The M-161( )/TNS-10 shall be subjected to the shock test on the "Shock Testing Machine for Light Weight Equipment" as shown in MIL-S-901 as specified on Drawing SC-A-883796 (see 3.8.4).

4.9.6 Fungus test. The M-161( )/TNS-10 shall be subjected to the test of Method 508.1, Procedure I, of MIL-STD-810 as specified on Drawing SC-A-883796 (see 3.8.5).

4.10 Visual and mechanical inspection. The microphone shall be examined for the defects listed in MIL-STD-252, and Table V, and inspected to verify conformance to those portions of 3.2.2 which can be determined visually.

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Table V. Classification of Visual and Mechanical Defects

Classification	Defects
Major	Interchangeability of parts Improper fit of cover Improper fit of acoustical plugs Batteries loose in holders Loose binding posts Loose plug connection Improper fit of hot-wire element Improper fit of lightning arrestors
Minor	Loose fasteners Fasteners difficult to operate Untidy wiring

4.11 Quality conformance inspection of packaging. Packaging shall be inspected in accordance with MIL-S-10956 to determine conformance to the requirements of Section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-10956.

6. NOTES

6.1 Intended use. Microphone M-161( )/TNS-10 is intended for use with Sound Ranging Set AN/TNS-10 for the detection of weapon signals.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Level of packaging and level of packing required for shipment (Level A, level B, or Commercial).

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c. Number of first articles to be submitted for approval (see 4.3.1).

d. Submission of the statement of treatment before first article approval. This statement should be submitted to the contracting officer (see 3.8.5).

e. A pistonphone box for use in performing the tests of Section 4 will be loaned to the contractor for the duration of the contract.

6.3 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: M-161R/TNS-10. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

6.4 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (see 4.5.3.3).

6.5 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-M-49123(EL) MICROPHONE M-161( )/TNS-10

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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